Recitation 4

1. (Chebyshev bound) Let $\tilde{x} \sim \text{Exponential}(\beta)$, $\mu_{\tilde{x}}$ the mean of $\tilde{x}$ and $\sigma_{\tilde{x}}^2$ the variance of $\tilde{x}$.
   (a) Compute $\mathbb{P}\left(|\tilde{x} - \mu_{\tilde{x}}| > k\sigma_{\tilde{x}}\right)$ for $k > 1$.
   (b) Compare this to the bound you get from Chebyshev's inequality.

   Solution. Let $F_{\tilde{x}}$ be the CDF of $\tilde{x}$. We have:

$$\begin{aligned}
\mathbb{P}\left(|\tilde{x} - \mu_{\tilde{x}}| > k\sigma_{\tilde{x}}\right) &= 1 - \mathbb{P}\left(-k\sigma_{\tilde{x}} < \tilde{x} - \mu_{\tilde{x}} < k\sigma_{\tilde{x}}\right) \\
&= 1 - \mathbb{P}\left(\mu_{\tilde{x}} - k\sigma_{\tilde{x}} < \tilde{x} < \mu_{\tilde{x}} + k\sigma_{\tilde{x}}\right) \\
&= 1 - F\left(\mu_{\tilde{x}} + k\sigma_{\tilde{x}}\right) + F\left(\mu_{\tilde{x}} - k\sigma_{\tilde{x}}\right) \\
&= 1 - 1 + \exp\left(-\frac{\beta + k\beta}{\beta}\right) + 0 \\
&= e^{-(1+k)}
\end{aligned}$$

   Chebyshev's bound provides, for $c = k\sigma_X$,

$$\mathbb{P}\left(|X - \mu_X| \geq k\sigma_X\right) \leq \frac{\sigma_X^2}{k^2\sigma_X^2} = \frac{1}{k^2}$$

   which is a weaker bound.

2. Let $\tilde{x}_1, \ldots, \tilde{x}_n \sim \text{Uniform}\ (0, \theta)$. Let $\tilde{y}_n = \max\left\{\tilde{x}_1, \ldots, \tilde{x}_n\right\}$.
   (a) Is $\tilde{y}$ an unbiased estimator of θ?
   (b) Is $\tilde{y}$ a consistent estimator of θ?

   Solution:

   (a) For any c, $\mathbb{P}(\tilde{y} < c) = \mathbb{P}\left(\tilde{x}_1 < c, \tilde{x}_2 < c, \ldots, \tilde{x}_n < c\right) = \mathbb{P}\left(\tilde{x}_1 < c\right)\mathbb{P}\left(\tilde{x}_2 < c\right)\ldots\mathbb{P}\left(\tilde{x}_n < c\right)$. The probability density function is

$$F_{\tilde{y}_n}(y) = \mathbb{P}(\tilde{y}_n < y) = \prod_{i=1}^{n} \mathbb{P}\left(\tilde{x}_i < y\right) = \left(\frac{y}{\theta}\right)^n$$

   for $0 \leq y \leq \theta$. The probability density function for $\tilde{y}$ is

$$f_{\tilde{y}_n}(y) = \frac{d}{dy} F_{\tilde{y}_n}(y) = \frac{n}{\theta^n} y^{n-1}$$

   The expectation of $\tilde{y}$ is

$$\mathbb{E}(\tilde{y}_n) = \int_0^\theta f_{\tilde{y}_n}(y) dy = \frac{n}{n+1}\theta < \theta$$

   Therefore, $\tilde{y}_n$ is a biased estimator.

   (b) For a given $\epsilon > 0$, we have

$$\mathbb{P}\left(\tilde{y}_n < \theta - \epsilon\right) = \prod_{i=1}^{n} \mathbb{P}\left(\tilde{x}_i < \theta - \epsilon\right) = \left(1 - \frac{\epsilon}{\theta}\right)^n$$

   which goes to 0 as $n \to \infty$, so $\lim_{n\to\infty} \hat{\theta}_n = \theta$, and thus $\tilde{y}_n$ is consistent.